Force sketching in MD: Can we replace multi-head committees with screening methods for faster force uncertainty compute?

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Abstract

Machine-learned interatomic potentials simulate molecules far more cheaply than quantum chemistry, but a simulator-based workflow only benefits if the emulator can say where it is misspecified — that is what decides when to fall back on the expensive mechanistic model. Committee disagreement is the usual signal, and a *multi-head* committee makes it nearly free for energies: M heads share one trunk, so one forward pass gives all M energies. Forces are not free. Forces are gradients, and each head's forces need their own backward pass, so force disagreement — the quantity acquisition actually uses — costs about M times more. We ask whether a handful of cheap random probes can estimate it instead, on a pretrained eight-head committee across four molecular systems. The answer is no for *replacement* and yes for *screening*. Even with six of the seven probes the exact computation would use, the best estimator recovers only 72–80% of the genuinely most uncertain structures. But a calibrated filter built from the same score safely skips 84–87% of the expensive computations while still catching 96–98%, beating cheaper alternatives in 11 of 12 comparisons. The wall-clock benefit is regime-dependent: $1.85\times$ in batch, only $1.11\times$ one structure at a time. The decision-quality result is robust; the speed result is not.

1 Introduction

Uncertainty quantification is what makes learned interatomic potentials usable beyond their training distribution: it drives active learning, flags extrapolation during molecular dynamics, and decides when a structure must go back to the quantum-chemistry calculation the model was trained to imitate, orders of magnitude slower and the very thing we are avoiding. Committee disagreement remains the most widely used estimator of that error: where training data pinned the model down its members agree, and where it did not they diverge, so their spread stands in for an error we cannot see [Behler, 2015, Podryabinkin and Shapeev, 2017, Carrete et al., 2023]. Multi-head committees [Beck et al., 2025] — shallow ensembles more broadly [Kellner and Ceriotti, 2024] — make that cheap by attaching M readout heads to one shared message-passing trunk, so all M energies come from a single forward pass instead of M separately trained models.

That amortization does not extend to forces. As the multi-head committee work states directly, "a single backward pass can not yield the forces for all the heads ... the standard deviation requires multiple reverse passes through the computational graph." Forces are what MD and most acquisition rules actually consume, so the expensive half of committee uncertainty is exactly the half the shared trunk fails to amortize.

This is a question about an imperfect simulator. A learned potential is an emulator of an expensive mechanistic model — density-functional theory — and is misspecified unevenly across configuration

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space. A simulator-based workflow does not need it right everywhere, only a trustworthy account of *where* it is not, so the expensive model is called exactly there. That multi-fidelity decision rests on committee disagreement; we ask how cheaply that signal can be had, and where it cannot.

Downstream, though, one rarely needs every member force: acquisition needs a *ranking*, a safety gate needs a *threshold decision*, error analysis needs the most uncertain *atom*. That motivates estimating the force-variance statistic directly rather than reconstructing the member forces defining it. We call this ForceSketch: replace the exact $(M-1)$-direction centered head basis with K randomized or orthogonal directions, each still costing one reverse pass.

2 Problem formulation

Setup and notation. A structure of N atoms has Cartesian coordinates $x \in \mathbb{R}^D$ with $D = 3N$. We index a single coordinate by $d \in \{1, \dots, D\}$, an atom by $a \in \{1, \dots, N\}$, and the three Cartesian components of one atom by $\gamma \in \{1, 2, 3\}$, so the pair (a, γ) names the same thing as d. The committee has M *heads*: M small readout networks attached to one shared message-passing trunk. Because they share the trunk, a single forward pass returns all M head energies at once, as the vector $E(x) = [E_1(x), \dots, E_M(x)]^\mathsf{T} \in \mathbb{R}^M$. Head m's force field is the negative energy gradient $f_m = -\nabla_x E_m \in \mathbb{R}^D$, and we collect the M of them as the columns of

$$F(x) \;=\; [\, f_1 \;\cdots\; f_M \,] \;\in\; \mathbb{R}^{D \times M}, \qquad \text{so } F_{dm} \text{ is head } m\text{'s force along coordinate } d.$$

Write $\bar{F}_d = \frac{1}{M} \sum_{m=1}^{M} F_{dm}$ for the committee-mean force on coordinate d — the force the simulation actually uses.

Two disagreement scores, at two scales. Uncertainty is how much the heads disagree. On a *single coordinate* that is their sample variance; summing over coordinates gives one number for the *whole structure*:

$$\underbrace{v_d}_{\text{per coordinate}} \;=\; \frac{1}{M-1} \sum_{m=1}^{M} \big(F_{dm} - \bar{F}_d\big)^2, \qquad \underbrace{S}_{\text{per structure}} \;=\; \sum_{d=1}^{D} v_d \;=\; \frac{\|FP\|_F^2}{M-1}. \tag{1}$$

Here $\mathbf{1} \in \mathbb{R}^M$ is the all-ones vector, $P = I_M - \frac{1}{M}\mathbf{1}\mathbf{1}^\mathsf{T}$ is the projector that subtracts the head mean (*centering*), and $\|\cdot\|_F$ is the Frobenius norm. Centering is what turns raw predictions into disagreement: FP holds each head's deviation from the consensus, and v_d is just the squared length of its d-th row, $v_d = \|(FP)_{d,:}\|_2^2/(M-1)$. Pooling the three components of one atom gives the two atom-level scores used in the literature, $u_a^{\mathrm{RMS}} = \big(\frac{1}{3}\sum_\gamma v_{a\gamma}\big)^{1/2}$ and $u_a^{\mathrm{MHC}} = \frac{1}{3}\sum_\gamma \sqrt{v_{a\gamma}}$, where $v_{a\gamma}$ is v_d for the coordinate $d = (a, \gamma)$. Section 4 says which of these the acquisition rule ranks by.

Why energies are cheap and forces are not. The M energies cost one forward pass because the heads share a trunk. Forces are gradients, and reverse mode returns the gradient of *one scalar* per backward pass — we call one such pass a *lane*. Since S depends on F only through FP, whose columns sum to zero, it lives in an $(M-1)$-dimensional centered subspace with orthonormal basis Q; differentiating the scalar $q_j^\mathsf{T} E$ costs one lane and returns $g_j = Fq_j$, with $v_d = \frac{1}{M-1}\sum_j g_{j,d}^2$. Exact force uncertainty therefore costs M lanes: $M-1$ centered directions plus the mean force MD needs anyway (Appendix D). *Our question is whether downstream decisions really need all $M-1$ of them.*

3 Method

Gaussian and Haar sketches. Our estimators are the molecular-UQ instance of randomized trace and norm estimation [Hutchinson, 1990, Avron and Toledo, 2011, Meyer et al., 2021]; the orthogonal variant is the head-space analogue of orthogonal random features [Yu et al., 2016], and the control variate plays the role that a low-rank sketch plays in randomized linear algebra [Halko et al., 2011]. Unbiased approximate vector–Jacobian products have also been used to cheapen *training* [Bakong et al., 2026], by sparsifying the backward pass inside the network; we instead compress the *output-head query space* at inference, leaving each reverse pass exact and reducing how many are needed to estimate an existing committee statistic. The idea is to stop asking for all $M-1$ centered directions

and instead *probe* the centered subspace with $K \ll M-1$ random ones, then rescale. Each probe still costs exactly one lane, so the question is how much of the disagreement K probes recover.

Concretely, draw $z_k \sim \mathcal{N}(0, I_M)$ and center it, $w_k = Pz_k$; one lane returns $g_k = Fw_k \in \mathbb{R}^D$, and

$$\widehat{v}_d \;=\; \frac{1}{K(M-1)} \sum_{k=1}^{K} g_{k,d}^2 \tag{2}$$

is an unbiased estimator of v_d — on average it gives the right answer, though any single draw does not. The intuition is the standard one behind randomized trace estimation: a random probe sees each direction of the subspace equally in expectation, so the average of K probes estimates the total energy in all $M-1$ of them.

Random probes can overlap, wasting budget. Sampling them *orthogonally* instead removes that waste. Let $O \in \mathbb{R}^{(M-1)\times K}$ have orthonormal columns drawn Haar-uniformly (that is, uniformly over all K-dimensional orientations, obtained in practice by orthonormalizing a Gaussian matrix), and set $W = QO \in \mathbb{R}^{M\times K}$: a random orthonormal K-frame lying inside the centered subspace. Collecting the K lanes it costs into $G = FW \in \mathbb{R}^{D\times K}$ — so G_{dk} is the response of coordinate d to probe k — gives

$$\widehat{v}_d^{\,\mathrm{Haar}} \;=\; \frac{1}{K} \sum_{k=1}^{K} G_{dk}^2, \tag{3}$$

which is unbiased and becomes *exact* at $K = M-1$, where the frame is a full basis of the centered subspace. Writing $r = M-1$ for its dimension, the two estimators have relative standard deviations $\sqrt{2/K}$ and $\sqrt{2(r-K)/(K(r+2))}$, so orthogonalization multiplies the estimator variance by exactly $(r-K)/(r+2)$ — a factor of $4/9$ at K=3, M=8. It costs nothing extra, and it is the single largest methodological lever we find.

Finite-K corrections. $\sqrt{\widehat{v}_d}$ is biased low, because the square root is concave. A multiplicative constant depending only on K (and, for orthogonal probes, on r) removes that bias; Appendix I gives both constants, including a form of the Haar constant that is finite at $K = r$ where the conventional Beta expression is singular.

Low-rank control variate. Disagreement is not spread evenly over the centered subspace, so probing it obliviously wastes budget. Split $P = Q_{r_0} Q_{r_0}^{\mathsf{T}} + R$, evaluate the r_0 leading directions of the design-split average $\bar{A} = \frac{1}{n}\sum_i PF_i^{\mathsf{T}} F_i P$ exactly, and sketch only the residual with $K' = K - r_0$ Haar probes. The estimator stays unbiased (Appendix E).

Screening rather than replacement. If the sketch cannot *replace* the exact score, it can still certify when that score is small. Split conformal [Vovk et al., 2005, Angelopoulos et al., 2024] turns any score into a calibrated bound from held-out data alone: on a calibration split form the ratios $\eta_i = S(x_i)/(\widehat{S}(x_i) + \epsilon)$ and let c_α be their $\lceil (n+1)(1-\alpha) \rceil$-th order statistic. Then $U(x) = c_\alpha \widehat{S}(x)$ sits above $S(x)$ at least $1-\alpha$ of the time, so $U(x) < \tau$ certifies the exact score is below the high-uncertainty threshold: skip when it is, fall back otherwise. This is *marginal* coverage under exchangeability, not a conditional or out-of-distribution guarantee (Appendix F).

4 Experimental setup

We use the pretrained eight-head MACE [Batatia et al., 2022] committees released with Beck et al. [2025]: two message-passing layers, 32 channels, $\ell_{\max} = 1$, $r_{\mathrm{cut}} = 6$ Å. We evaluate the three head-distribution variants (*disjoint*, *overlapping*, *same*) on 3BPA [Kovács et al., 2021] at 1200 K (2139 structures, extrapolative relative to the 300 K training set) and the joint ten-molecule rMD17 [Christensen and von Lilienfeld, 2020] committee on ethanol (9 atoms), aspirin (21) and azobenzene (24), 1000 held-out structures each. Estimator evaluation is in float64 and this is not fastidiousness: the mean force exceeds the disagreement by a factor $\kappa \approx 36$ here, so a float32 "exact" reference carries more relative error than the 10^{-5} tolerance at which we verify full-rank exactness. Deployment timing is float32, on one RTX 5070 Laptop GPU, with PyTorch [Paszke et al., 2019] CUDA events measured per iteration; the fitted lane cost is $T(L) = 10.38 + 11.78\,L$ ms. Ten random

seeds are frozen before testing and none discarded; all intervals are paired bootstraps [Efron and Tibshirani, 1993] over structures with 10,000 resamples.

The acquisition decision. Following Beck et al. [2025], structures are ranked by the maximum disagreement of their force components. We take the max over components, $\widehat{u}^{\max\text{-comp}} = \max_d \widehat{\sigma}_d$, as primary and the global sum $\widehat{S} = \sum_d \widehat{v}_d$ as an easier secondary; a hat always denotes the sketched estimate of the corresponding exact quantity. Section 5.1 measures top-5% recall against the exact score; Section 5.3 applies a fixed threshold τ transferred from the design split, so its realised test prevalence is 0.028–0.050. Every metric is computed per seed and then averaged, so the deployed budget is K lanes, not $10K$ (Appendix G).

Each system is split once into three disjoint parts — *design* (20%, fitting Q_{r_0} and τ), *calibration* (20%, fitting c_α alone) and *test* (60%, touched once). Split-conformal coverage requires the score function to be frozen independently of the calibration observations, so a basis fitted on calibration structures would void the guarantee (Appendix J).

5 Results

5.1 Can force disagreement be sketched?

Partially, and not well enough to replace exact computation — under either acquisition rule, and markedly less well under the primary one.

Under $u^{\max\text{-comp}}$, Haar is the best sketch at every budget, but its top-5% recall is only 0.362–0.489 at $K{=}3$ and 0.484–0.570 at $K{=}4$ across the 4 systems. Rank correlation is far healthier than tail selection (Spearman 0.648–0.781 at $K{=}4$), which is the recurring pattern: a sketch that orders structures tolerably still fails to identify *which* ones land in the extreme tail. On the easier global statistic the same estimator reaches 0.529 [0.490, 0.566] and 0.619 [0.580, 0.648] at $K{=}3, 4$ on 3BPA — roughly ten points higher, and still far from substitution quality; Appendix N plots that easier view against K alongside the latency curve. Intervals are 95% paired bootstraps over structures (10,000 resamples) on the seed-averaged statistic; seed-to-seed standard deviations are 0.02–0.09 and are recorded separately.

The extreme-value rule is harder for a reason we measure: a maximum over $3N$ noisy estimates is biased upward by 1.07–1.08 at $K{=}4$, which no marginal finite-K correction removes. That obstructs replacement but not screening, since a scale-calibrated gate absorbs a monotone bias into c_α (Appendix G).

Does sketching beat simply using fewer heads? It depends on whether an exact mean force is required, and the two framings give opposite answers, so Appendix R reports both rather than settling the comparison by definition.

5.2 Does reducing directions reduce GPU time?

Only in some regimes. Against a *batched* exact baseline — which this stack makes available only after disabling the TensorExpr fuser (Appendix M) — the total force-plus-UQ speedup at $K{=}3$ is $1.85\times$ at $B{=}16$ but collapses to $1.11\times$ at $B{=}1$, where extra cotangent lanes are nearly free. Appendix K gives the full accounting, the latency table and a kernel profile.

5.3 Can approximate screening avoid exact computation?

This is where the method succeeds — and, unlike Section 5.1, it succeeds under the primary max-component rule. With a control variate at $r_0{=}2$, $K{=}4$ and $\alpha{=}0.05$, evaluated on the held-out test split, with Q_{r_0} and τ fitted on the design split and c_α on the calibration split:

system	high-UQ recall	exact skipped	screening speedup
3BPA @ 1200 K	0.972 [0.949, 0.991]	0.844 [0.828, 0.860]	$1.41\times$
rMD17 ethanol	0.982 [0.940, 1.000]	0.836 [0.812, 0.859]	$1.40\times$
rMD17 aspirin	0.973 [0.946, 0.994]	0.851 [0.828, 0.873]	$1.41\times$
rMD17 azobenzene	0.964 [0.926, 0.992]	0.872 [0.849, 0.894]	$1.43\times$

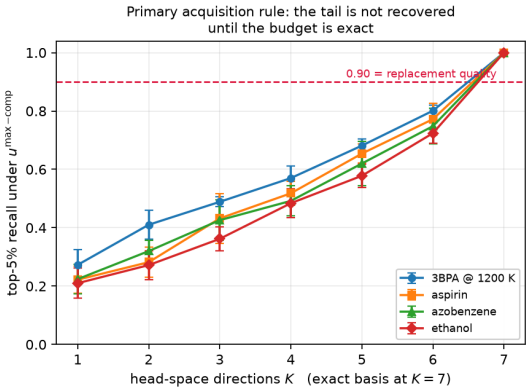


Figure 1: The negative result, under the *primary* max-component rule: top-5% recall against budget K on all four systems. Recall climbs steadily but never reaches the 0.90 replacement line short of the exact basis at $K{=}7$. Error bars are the standard deviation over ten frozen seeds. Appendix B plots the corresponding screening Pareto.

Between 0.836 and 0.872 of exact force-UQ evaluations are avoided while 0.964–0.982 of high-uncertainty configurations are still routed to the exact computation, on every system. Intervals are 95% paired bootstraps over test structures (10,000 resamples); the positive sets hold only tens of structures, so the recall intervals are correspondingly wide and we report them rather than the point estimates alone. The screening gate is where the extreme-value bias of Section 5.1 stops mattering: c_α is a multiplicative calibration constant, so any bias that is monotone in the estimator's noise is absorbed rather than propagated.

Every gate spends $1{+}K$ lanes per structure and, on fallback, only $r{-}K$ further passes rather than a fresh M — charged identically for all gates, since crediting that reuse to one method alone would manufacture its advantage (Appendix L).

Is the gate doing the work, or is ForceSketch? A calibrated gate is a generic wrapper, so the claim only stands if this gate beats the cheaper ones a reader will reach for. Against a *free* energy-disagreement gate, head subsampling with the exact mean, and a plain Haar sketch — all at matched lane budget, identical splits and threshold — the control variate *Pareto-dominates* in 11 of 12 gate-by-system comparisons. The free gate is the informative failure: its recall collapses to 0.636 on azobenzene, so energy disagreement is a genuinely different signal from force disagreement, not a cheap proxy for it. Appendix A gives the full comparison.

6 Conclusion and limitations

Exact multi-head force uncertainty cannot be replaced by a handful of randomized head-space probes: the best sketch reaches only 0.484–0.570 top-5% recall at $K{=}4$ and 0.724–0.801 even at $K{=}6$ of 7. It can be *screened*: a split-conformal gate over a control-variate sketch avoids 0.836–0.872 of exact computations while retaining 0.964–0.982 of high-uncertainty structures on every system — a property of the estimator, not the implementation. **Limitations.** Whether it saves *time* depends on the regime: at $B{=}1$ extra cotangent lanes are nearly free, the speedup falls to $1.11\times$ and the gate becomes a net slowdown, so this suits batched active-learning sweeps, not single-structure MD. We preserve a *proxy*, not a demonstrated outcome: no active-learning loop is run and no reference-force error measured, so we claim no gain in model quality or oracle-query efficiency. Committee disagreement is only a proxy for error; calibration is marginal and under exchangeability; $M{=}8$ bounds the maximum saving; and rMD17 spans only 9–24 atoms.

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A Screening-gate baselines at matched budget

Is the gate doing the work, or is ForceSketch? A calibrated gate is a generic wrapper — any score correlated with exact force disagreement can be calibrated the same way — so the claim only stands if this gate beats the cheaper ones a reader will reach for. We build three at matched lane budget with identical splits, calibration and threshold. The first is *free*: the M head energies come from the forward pass the model already runs, costing zero *uncertainty* lanes. The second is head subsampling using the *exact* mean force ForceSketch already pays a lane for, $\hat{v}_d = \frac{M}{K(M-1)} \sum_{i \in \mathcal{S}} (F_{di} - \bar{F}_d)^2$, unbiased under uniform sampling and markedly lower-variance than the sample variance among the drawn heads.

gate	lanes	exact skipped	high-UQ recall	speedup
energy disagreement (free)	1	0.294–0.525	0.636–1.000	1.05–1.33×
head subsampling, exact mean	5	0.474–0.747	0.847–0.936	1.16–1.33×
Haar sketch	5	0.525–0.734	0.870–0.926	1.19–1.32×
control variate	5	**0.836–0.872**	**0.964–0.982**	**1.40–1.43×**

Ranges are across the four systems. The control variate *Pareto-dominates* — strictly more skipped *and* strictly higher recall — in 11 of the 12 gate-by-system comparisons, including every one against the two sketch baselines. The exception: on ethanol the free energy gate attains perfect recall (1.000) while skipping only 0.322. Elsewhere that free gate is the informative failure, collapsing to 0.636 recall on azobenzene: energy disagreement is a genuinely different signal from force disagreement, not a cheap proxy for it.

B Screening Pareto at matched budget

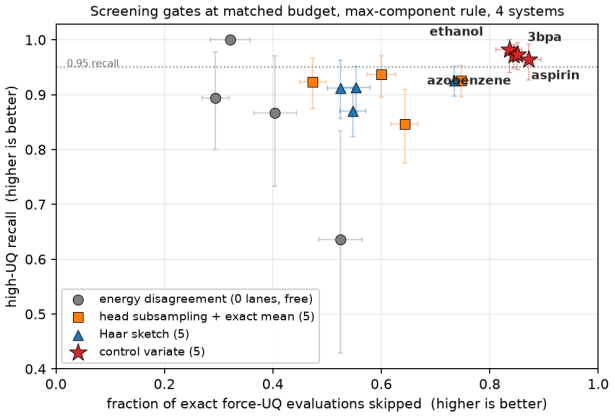


Figure 2: All four gates at matched lane budget on all four systems, under the max-component rule. Up-and-right is better on both axes, so the control variate's dominance is read directly. Bars are 95% paired bootstrap intervals.

C Data and code availability

All datasets and model checkpoints used here are public and are pinned by SHA-256 in the artifact's freeze manifest: the 3BPA [Kovács et al., 2021] and rMD17 [Christensen and von Lilienfeld, 2020] datasets, and the pretrained multi-head committees released with Beck et al. [2025] (Zenodo record 17829635). The estimator library, every experiment script, the analysis code, the ten frozen random seeds and every raw result record behind the numbers in this paper are available at `https://github.com/extremefattypunch/forcesketch`. Everything downstream of the cached head-force matrix is offline linear algebra and reproduces without a GPU; only the timing experiments require one.

D Lanes and the centered head subspace

Why energies are cheap and forces are not. The M energies cost one forward pass because the heads share a trunk. Forces are gradients, and reverse-mode differentiation returns the gradient of *one scalar* per backward pass. Feeding it the mean energy yields the mean force, not the spread; feeding it each E_m in turn yields all M force fields but costs M backward passes. We call one backward pass a *lane*.

The saving grace is that S depends on F only through the centered matrix FP, whose columns sum to zero and therefore span at most $M-1$ dimensions. Let $Q \in \mathbb{R}^{M\times(M-1)}$ have orthonormal columns $q_1, \ldots, q_{M-1}$ spanning exactly that centered subspace:

$$Q^\mathsf{T}Q = I_{M-1}, \qquad Q^\mathsf{T}\mathbf{1} = 0, \qquad QQ^\mathsf{T} = P.$$

Each column is one lane's worth of work. Differentiating the scalar $q_j^\mathsf{T}E$ — a fixed weighted combination of the head energies — costs a single backward pass and returns

$$g_j \;=\; -\nabla_x\big(q_j^\mathsf{T}E\big) \;=\; Fq_j \;\in\; \mathbb{R}^D, \qquad \text{and} \qquad v_d \;=\; \frac{1}{M-1}\sum_{j=1}^{M-1} g_{j,d}^2. \tag{4}$$

So the exact variance needs $M-1$ lanes, plus one more along the uniform direction $\frac{1}{M}\mathbf{1}$ for the mean force the simulation needs anyway: M lanes in total, matching the naive per-head loop. *Our question is whether downstream uncertainty decisions really need all $M-1$ centered lanes.*

E The low-rank control variate

Low-rank control variate. Split $P = Q_{r_0}Q_{r_0}^\mathsf{T} + R$, where $Q_{r_0} \in \mathbb{R}^{M\times r_0}$ has orthonormal columns inside the centered subspace and R is the projector onto what is left of it, the *residual* subspace of dimension $r - r_0$. The columns of Q_{r_0} are the r_0 leading eigenvectors of the *design*-split average $\bar{A} = \frac{1}{n}\sum_i PF_i^\mathsf{T}F_iP$ — an $M \times M$ matrix in *head* space, whose top eigenvector is the weighting of heads along which this committee disagrees most consistently. It is fitted separately for each system on structures disjoint from both the calibration and the test splits (Section 4). Fitting it on the calibration split instead would invalidate the conformal guarantee below, since split conformal requires the score function — learned basis included — to be frozen before the calibration ratios are formed. The idea is to spend a few lanes where the disagreement actually concentrates, and probe only what is left over. Evaluate those r_0 leading directions exactly, costing $g_j = Fq_j$ for the columns q_j of Q_{r_0}, and sketch only the residual with $K' = K - r_0$ Haar probes drawn inside R's range, collected as $G \in \mathbb{R}^{D\times K'}$ exactly as before but with the frame now living in the residual subspace:

$$\widehat{v}_d \;=\; \frac{1}{r}\Big[\textstyle\sum_{j\le r_0} g_{j,d}^2 \;+\; \frac{r-r_0}{K'}\sum_{k=1}^{K'} G_{dk}^2\Big], \tag{5}$$

which remains unbiased. By *spectrum* we mean the eigenvalues of $\bar{A}$ — how the total disagreement is divided among head-space directions. The motivation is then the *opposite* of the intuitive one: a concentrated spectrum does not explain why small K works, it is what *breaks* a random sketch, since a probe either catches the dominant direction or misses it.

F Split-conformal screening

Screening rather than replacement. If the sketch cannot *replace* the exact score, it can still say when the exact score is certainly small. Conformal prediction [Vovk et al., 2005, Angelopoulos et al., 2024] turns any score, however crude, into a calibrated bound using held-out data alone. On a calibration split of n structures disjoint from design and test, form the ratios $\eta_i = S(x_i)/(\widehat{S}(x_i) + \epsilon)$, the factor by which the cheap score underestimates the exact one — and let c_α be their $\lceil (n+1)(1-\alpha)\rceil$-th order statistic, large enough to cover all but an α fraction of them. Then $U(x) = c_\alpha\widehat{S}(x)$ sits above $S(x)$ at least $1-\alpha$ of the time, so $U(x) < \tau$ *certifies* that the exact score is below the high-uncertainty threshold τ too: skip when $U(x) < \tau$, fall back otherwise. Formally this is *marginal* coverage, $\mathbb{P}[S(x) \le U(x)] \ge 1-\alpha$, under exchangeability — not a conditional guarantee and not an out-of-distribution one. 3BPA at 1200 K calibrated against 300 K behaviour is exactly such a shift, so we report the observed coverage there rather than assume it.

G The acquisition decision and the extreme-value bias

The acquisition decision, stated exactly. Every number below concerns one decision: from a pool of held-out structures, select the most force-uncertain fraction for exact reference computation. Following Beck et al. [2025], structures are ranked by the maximum disagreement of their force components; we evaluate that extreme-value rule in both of its defensible forms — maximum over components, maximum over atoms — and report the global sum as an easier diagnostic statistic:

$$\widehat{u}^{\max\text{-comp}}(x) = \max_d \widehat{\sigma}_d, \qquad \widehat{u}^{\max\text{-atom}}(x) = \max_a \widehat{u}_a^{\mathrm{MHC}}, \qquad \widehat{S}(x) = \sum_d \widehat{v}_d. \tag{6}$$

A hat always denotes the sketched estimate of the corresponding exact quantity: the exact scores are the same three expressions with v_d and $\sigma_d = \sqrt{v_d}$ in place of $\widehat{v}_d$ and $\widehat{\sigma}_d$, so the exact S here is exactly the S of Eq. (1). The correction c is c_K for Gaussian probes and $c_{K,r}^{\mathrm{Haar}}$ for orthogonal ones (Section 3); u_a^{MHC} is as in Section 2. We take $u^{\max\text{-comp}}$ as primary, being the most literal reading, report $u^{\max\text{-atom}}$ alongside it in the artifact, and treat S as the statistically *easier* secondary: a sum over $3N$ coordinates concentrates, whereas both maxima are extreme-value statistics of a noisy $\widehat{\sigma}$.

The two experiments select positives differently and we do not conflate them. Section 5.1 measures **top-5% recall**: the 5% of evaluated structures with the largest *exact* score, ties resolved inclusively, so the set may slightly exceed 5%. Section 5.3 applies a *fixed* threshold instead, as a deployed gate must: τ is the 95th percentile of the exact score on the design split, and a test structure counts as high-uncertainty when its exact score is at least τ — so the realised test prevalence is 0.028–0.050, not exactly 0.05. Every metric is computed *per seed and then averaged* over the ten frozen seeds: the deployed budget is K lanes, not $10K$, and averaging the sketches before scoring would be a different, far more accurate estimator than the one we claim.

Why the extreme-value rule is harder. Making each $\widehat{\sigma}_d$ unbiased *marginally* is not the same as making $\max_d \widehat{\sigma}_d$ unbiased: a maximum over $3N$ noisy estimates is biased upward, and the bias grows with the estimator's noise. Measured directly, $u^{\max\text{-comp}}$ is inflated by 1.42–1.53 at $K{=}1$, falling to 1.07–1.08 at $K{=}4$ and to 1.000 at the full basis, where the estimator is exact and the bias must vanish — a useful check on the pipeline. This obstructs *replacement* but not screening (Section 5.3): the bias is monotone in the estimator's noise, so a scale-calibrated gate absorbs it into c_α.

Orthogonalizing the probes is the largest single lever in the method. The estimator comparisons that follow are on the global statistic, where the paired-bootstrap machinery of Section 4 was frozen; the ordering they establish is a property of the estimators, not of the acquisition rule. The paired difference between Haar and Gaussian at $K{=}3$, evaluated on identical structure resamples, is 0.225 [0.186, 0.263] in top-5% recall, and is significant on all four systems. The low-rank control variate adds a further 0.111 [0.080, 0.145] over Haar at $K{=}4$, reaching 0.730 [0.692, 0.760] — still short of 0.90.

The head-space spectrum explains why. Writing $A = Q^{\mathsf{T}} \bar{F}^{\mathsf{T}} F Q$ for the averaged centered force-disagreement Gram matrix, the stable rank of the matrix we actually sketch is $\mathrm{srank}(FQ) = \|FQ\|_F^2 / \|FQ\|_2^2 = \mathrm{tr}(A)/\lambda_{\max} = 2.91$–$3.08$ of a maximum 7 across the four systems, with the leading direction carrying 0.325–0.344 of the disagreement (an isotropic spectrum would give 0.143). The spectrum is therefore *moderately diffuse rather than strongly rank-concentrated* — but far from isotropic, at 2.3–$2.4\times$ the isotropic leading share. That is exactly the regime in which a random K-dimensional projection pays a variance penalty, and it is why projecting out the leading directions (the control variate) recovers so much.

H What ForceSketch replaces

I Finite-K corrections

Finite-K corrections. Reporting a standard deviation means taking a square root, and the square root of an unbiased variance estimate is *not* an unbiased standard-deviation estimate: $\sqrt{\cdot}$ is concave, so $\sqrt{\widehat{v}_d}$ is biased low, the more so the fewer probes we spend. The fix is a multiplicative constant depending only on K (and, for orthogonal probes, on r). For Gaussian probes $K\widehat{v}_d / v_d \sim \chi^2_K$ and the correction is $c_K = \sqrt{2/K}\,\Gamma(\frac{K+1}{2})/\Gamma(\frac{K}{2})$. For Haar probes the analogous constant is conventionally

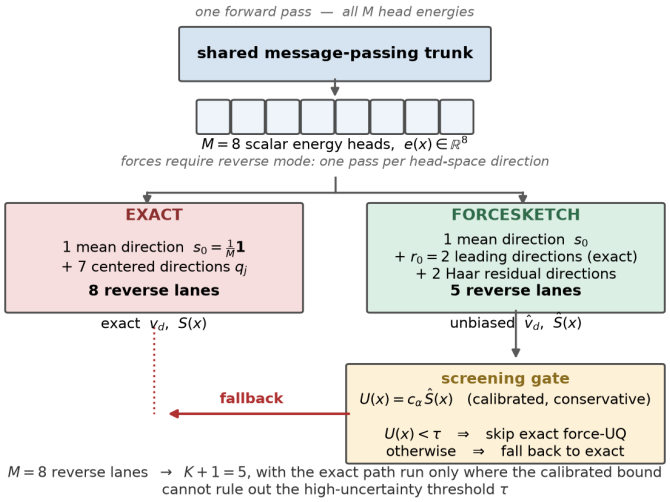


Figure 3: What ForceSketch replaces. All M head energies come from one forward pass, but forces do not: exact disagreement needs the full centered head subspace, M lanes. ForceSketch spends $K{+}1$, and — since that is not accurate enough to *replace* exact uncertainty at $K \leq 4$ under either acquisition rule (Section 5.1) — its score feeds a calibrated gate that runs the exact path only where the bound cannot rule out the threshold. Whether fewer lanes also cost less time depends on the autodiff regime (Section K.1).

written with Beta functions, which is singular at $K = r$ (it evaluates $B(K/2, 0)$). Cancelling the shared $\Gamma(\frac{r-K}{2})$ gives a form that is finite everywhere and exactly 1 at full rank:

$$c_{K,r}^{\mathrm{Haar}} \;=\; \sqrt{\frac{r}{K}}\,\frac{\Gamma\left(\frac{K+1}{2}\right)\Gamma\left(\frac{r}{2}\right)}{\Gamma\left(\frac{K}{2}\right)\Gamma\left(\frac{r+1}{2}\right)}, \qquad c_{K,r}^{\mathrm{Haar}} \to c_K \;\text{ as } r \to \infty. \tag{7}$$

We verify unbiasedness of $\sqrt{\widehat{v}}/c$ to within Monte-Carlo error at 4×10^5 draws. These closed forms cover the Gaussian and Haar estimators; the control variate below mixes exact and sketched directions and has no such constant, so we apply the residual frame's Haar correction, which is an approximation. It does not affect the screening results, because c_α is itself a multiplicative calibration and absorbs any constant scale error exactly.

J Splits

Splits. Each system is partitioned once, under a frozen permutation, into three disjoint splits: a *design* split (20%) fitting the control-variate basis Q_{r_0} and the threshold τ; a *calibration* split (20%) fitting the conformal constant c_α alone; and a *test* split (60%), touched once — $427/428/1284$ structures for 3BPA and $200/200/600$ per rMD17 molecule. The separation is not cosmetic: split-conformal coverage requires the score function to be frozen independently of the calibration observations, so a basis fitted on calibration structures would void the guarantee. The code asserts all three pairwise disjointness conditions and that the splits partition the set.

K Does reducing directions reduce GPU time?

K.1 Does reducing directions reduce GPU time?

Only in some regimes, and the honest answer required first making the strongest baseline available. Batched reverse mode — vectorizing the backward over cotangents while reusing one forward graph — is the baseline fairness demands, and on this stack it fails from the third call onwards, which disguises the cause. Appendix M diagnoses it (the TensorExpr fuser and e3nn's scripted spherical harmonics [Geiger and Smidt, 2022]); Appendix P shows that compilation does not compose with it; without the fix we would have compared against a Python loop and overstated every speedup below.

Two ratios answer different questions and we never conflate them. With $T(L)$ the latency of an L-lane reverse pass, the *incremental UQ* speedup $[T(M) - T(1)]/[T(1{+}K) - T(1)]$ is the cost of uncertainty once the mean force is paid for, while the *total force-plus-UQ* speedup $T(M)/T(1{+}K)$ includes it. We lead with the total, because the incremental ratio is unstable at small batch precisely when $T(1{+}K) \approx T(1)$ — which is the effect this section is about.

That effect is large. Batched reverse mode is nearly *flat* in the lane count at $B{=}1$ (25.5 ms for one lane, 30.5 ms for eight), so extra cotangents are almost free and the total speedup at $K{=}3$ collapses to $1.11\times$ — against $1.83\times$ if one compares only to a serial loop. At $B{=}16$ the batched path scales with L again and the speedup recovers to $1.85\times$; by $B{=}64$ it is slower than the serial loop at *every* lane count measured ($1.22\times$ at $L{=}8$), so serial becomes the strongest baseline again and the speedup is $1.86\times$. Incremental speedups are $1.63\times$ and $3.18\times$ at $B{=}1$ and $B{=}16$. Appendix O tabulates the latencies behind these ratios and Appendix Q profiles where the GPU time goes.

L What a screened evaluation costs

What a screened evaluation costs. Every gate spends $1{+}K$ lanes per structure: one mean-force lane, which MD needs regardless, plus K uncertainty lanes. On fallback it does *not* restart: the K directions already evaluated are linearly independent inside the r-dimensional centered subspace, and an orthonormal frame spanning them is a recombination of vectors already in hand, so completing the exact basis costs only $r{-}K$ further passes. With p the fallback fraction, $T_{\mathrm{screen}} = T(1{+}K) + pT(r{-}K)$ against $T_{\mathrm{exact}} = T(M)$ — at $K{=}4$, $M{=}8$, $T(5) + pT(3)$, while the free energy gate, evaluating no centered direction, pays $T(1) + pT(7)$. Crediting that reuse to the control variate alone would manufacture much of its advantage, so every gate is charged this way. $T(\cdot)$ is the measured median at $B{=}16$ under whichever implementation is faster at that lane count (Section K.1); fallback is charged a fresh forward rather than a retained graph, which is conservative; and the quoted figure is the *total* force-plus-UQ speedup.

M Diagnosing the batched reverse-mode failure

Batched reverse mode, and why it initially appeared impossible. `torch.autograd.grad(is_grads_batched=True)` vectorizes the backward over cotangents while reusing one forward graph, and is the baseline spec-level fairness demands. On this stack it fails with `Cannot access data pointer of Tensor that doesn't have storage` — but only from the third call onwards, which disguises the cause. The mechanism is: batched reverse mode runs under `torch._vmap_internals`, so cotangents are `BatchedTensors` with no storage; TorchScript's profiling executor emits an optimized plan only after two warm-ups; and the TensorExpr fuser then fuses the *reverse* graph into a `prim::TensorExprGroup`, whose kernel requires a raw `data_ptr()` for every input. The TorchScript on the path is `e3nn`'s `_spherical_harmonics`, a module-level `@torch.jit.script` *free function* — not a module, so walking the module tree finds nothing to un-script, and a fully rebuilt zero-`ScriptModule` model still fails. Disabling the TensorExpr fuser before the first forward fixes it: 50/50 consecutive batched calls agreeing with the serial path to 3×10^{-15}, at a $\sim 4\%$ cost to the serial path from losing fusion. We report this because it is a trap any e3nn-based UQ work will hit, and because without it we would have compared against a Python loop, which spec-level fairness forbids.

N Accuracy and cost on the global statistic

Figure 1 (left) plots recall against K under the primary max-component rule. For completeness the same trade-off on the easier global statistic S, together with the latency curve, is below; it is the secondary view, and is uniformly more optimistic.

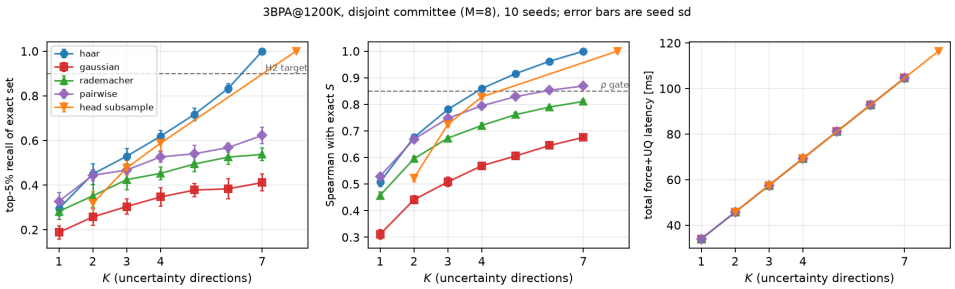


Figure 4: Accuracy and cost against K, ranked by the global statistic S. Latency curves coincide for every method at equal lane count under the serial implementation; batched reverse mode flattens them (Section K.1). Ranking saturates well before tail selection: Haar clears Spearman $\rho = 0.85$ at K=4 while its top-5% recall is still 0.619. Error bars are the standard deviation across the ten frozen seeds.

O Measured serial and batched latency

	$B = 1$ (27 atoms)		$B = 16$ (432 atoms)	
lanes L	serial	batched	serial	batched
1	**22.9**	25.5	**24.3**	26.0
4	60.3	**27.5**	62.5	**39.7**
8	110.5	**30.5**	115.5	**73.2**

Median latency in ms, fp32, one RTX 5070 Laptop GPU, CUDA events with one event pair per iteration; **bold** marks the faster implementation at that (B, L). These are the measurements behind every speedup quoted in Section K.1.

P Compilation, and why it does not compose

For completeness we also compiled the model. `torch.compile` [Ansel et al., 2024] succeeds only partially here — Dynamo builds a sequence-length guard on e3nn's `Irrep` and calls `len()` on it, which e3nn raises `NotImplementedError` for by design, so untraceable regions must fall back to eager. With that, compilation is correct to 2.2×10^{-7} and gives 1.04–$1.10\times$ at B=16. The absolute saving grows linearly in L, so AOTAutograd is producing a cheaper backward rather than merely a faster forward.

The obvious next question is whether the two accelerations *compose*: compiled-plus-batched would then be the baseline fairness demands. They do not. With the TensorExpr fuser already disabled, compiling the model and then taking a batched VJP reproduces the identical `Cannot access data pointer of Tensor that doesn't have storage` failure, at both B=1 and B=16 — AOTAutograd's compiled backward is no more able to accept a storage-free `BatchedTensor` than the fused TorchScript one was. We therefore report the eager batched path as primary because it is the fastest *correct* baseline available on this stack, not merely the most convenient one, and note that compilation alone would shift these numbers by at most a few percent in the same direction for every method, leaving the crossover of Section K.1 unchanged.

Q Where the GPU time goes

A kernel-level profile of the serial path explains why removing lanes helps at all when it does: the composition of GPU time is invariant in L (elementwise 42.2% at L=7 against 41.5–41.9% at L=2, 3, 4; tensor products 21.3% against 20.0–20.9%; the same 134 distinct kernels throughout), and only the repetition count changes: kernel launches fall by $2.12\times$ from the exact path to K=3, against a lane ratio of $2.33\times$. Sketching linearly reduces repeated execution of the shared trunk, and batched reverse mode attacks the same redundancy by a different route — which is exactly why the two are substitutive at small batch.

R Sketching versus using fewer heads

At equal total lanes, does a random sketch beat simply computing a few of the head forces exactly? The comparison hinges on which head-subsampling estimator is used, and the fair one is not the obvious one. Bare head subsampling takes the sample variance among the K drawn heads. But at matched budget it also pays a mean-force lane, and can therefore use the *exact* mean:

$$\widehat{v}_d^{\,\mathrm{sub}} \;=\; \frac{M}{K(M-1)} \sum_{i \in \mathcal{S}} \big(F_{di} - \bar{F}_d\big)^2,$$

unbiased under uniform sampling without replacement and markedly lower-variance than the sample variance, because it does not spend degrees of freedom re-estimating a mean already in hand.

Against that fair baseline the ordering *reverses*. Haar at $K{=}3$ scores -0.040 [-0.074, -0.008] against exact-mean head subsampling at $K{=}3$ on 3BPA and -0.070 [-0.126, -0.016] on aspirin — both significantly *negative*, i.e. head subsampling wins — and the difference is not significant on ethanol or azobenzene. An earlier version of this comparison scored the baseline with the sample variance while still charging it the mean lane, and reported Haar ahead; that was an artifact of hobbling the baseline, and we report the corrected result instead. Head subsampling at $K{=}4$ with no mean lane also beats Haar at equal lanes (-0.059 [-0.093, -0.023] on 3BPA), but cannot produce the mean force MD requires.

The paper's estimator-design claims do not rest on this comparison. Orthogonalization over Gaussian probes (0.225 [0.186, 0.263], significant on all four systems) and the control variate over Haar (0.111 [0.080, 0.145]) are unaffected, and it is the control variate — not the oblivious sketch — that wins the screening comparison of Section 5.3, where exact-mean head subsampling is included as a gate baseline in its fair form.